4.2 Isosceles & Equilateral Triangle Theorems

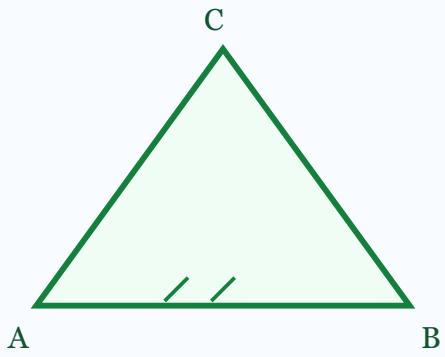
The **Base Angles Theorem** states that if two sides of a triangle are congruent, the angles opposite those sides (the base angles) are congruent. Its **converse** is also true: if two angles of a triangle are congruent, the sides opposite them are congruent. An equilateral triangle is a special case where all three sides are congruent; applying the Base Angles Theorem to each pair of equal sides shows all three angles must also be congruent, and since they sum to $180°$, each equals exactly $60°$ (the Equilateral Triangle Corollary). Conversely, an equiangular triangle must be equilateral. This worksheet applies these theorems with numeric problems, algebra, coordinates, and a short proof.

Examples

Example 1: Base angles from vertex angle

Isosceles $\triangle ABC$ has $AB = AC$ and vertex angle $\angle A = 40°$. Find each base angle.

Solution



$AB = AC$ (tick marks), so $\angle B \cong \angle C$ by the Base Angles Theorem.

Identify: $\angle B$ and $\angle C$ are the base angles (opposite the congruent legs AB and AC), so $\angle B = \angle C$.

Apply the Angle-Sum Theorem: $\angle A + \angle B + \angle C = 180$

$40 + 2\angle B = 180$

$2\angle B = 140 \Rightarrow \angle B = 70°$.

Answer: $\angle B = \angle C = 70°$.

Example 2: Vertex angle from base angles

Isosceles $\triangle ABC$ has base angles 55° each. Find the vertex angle.

Solution

Identify: the two given 55° angles are the base angles; the vertex angle is the remaining one.

Apply the Angle-Sum Theorem: vertex $= 180 - 55 - 55$

$= 180 - 110 = 70^\circ$.

Check: $55 + 55 + 70 = 180$ ✓.

Answer: Vertex angle $= 70^\circ$.

Example 3: Converse of the Base Angles Theorem

In $\triangle ABC$, $\angle B \cong \angle C$. What can you conclude about $\overline{AB}$ and $\overline{AC}$? Justify.

Solution

Identify: $\angle B$ and $\angle C$ are congruent angles.

The sides opposite these angles are $\overline{AC}$ (opposite $\angle B$) and $\overline{AB}$ (opposite $\angle C$).

By the **Converse of the Base Angles Theorem**, congruent angles imply the sides opposite them are congruent.

Answer: $\overline{AB} \cong \overline{AC}$ (so $\triangle ABC$ is isosceles with vertex A).

Example 4: Prove the Equilateral Triangle Corollary

Prove: if a triangle is equilateral, each of its angles measures 60°.

Solution

Let $\triangle ABC$ have $AB = BC = CA$ (equilateral, given).

Since $AB = AC$, the Base Angles Theorem gives $\angle B \cong \angle C$.

Since $BA = BC$, the Base Angles Theorem gives $\angle A \cong \angle C$.

Combining, $\angle A = \angle B = \angle C$. Let each equal x.

By the Angle-Sum Theorem: $x + x + x = 180 \Rightarrow 3x = 180 \Rightarrow x = 60$. ■

Conclusion: every angle of an equilateral triangle measures exactly 60°.

Example 5: Algebra — congruent base angles

Isosceles triangle has base angles $2x + 10$ and $3x - 5$. Find x, the base angles, and the vertex angle.

Solution

Identify: base angles are congruent, so set the expressions equal.

$2x + 10 = 3x - 5$

$15 = x$, so $x = 15$.

Each base angle $= 2(15) + 10 = 40°$ (check other expression: $3(15) - 5 = 40$ ✓).

Vertex angle $= 180 - 40 - 40 = 100°$.

Answer: $x = 15$; base angles $40°$ each; vertex angle $100°$.

Example 6: Algebra — vertex and base together

Isosceles triangle has vertex angle $3x$ and base angles $x + 30$ each. Find x and all three angles.

Solution

Set up the angle-sum equation using both base angles and the vertex angle:

$3x + (x + 30) + (x + 30) = 180$

$5x + 60 = 180$

$5x = 120 \Rightarrow x = 24$.

Vertex $= 3(24) = 72°$; base angles $= 24 + 30 = 54°$ each.

Check: $72 + 54 + 54 = 180$ ✓.

Answer: $x = 24$; vertex $= 72°$; base angles $= 54°$ each.

Example 7: Equilateral side algebra and perimeter

An equilateral triangle has two sides represented by $4x - 3$ and $2x + 7$. Find x, the side length, and the perimeter.

Solution

Since the triangle is equilateral, all sides are equal, so set the two expressions equal:

$4x - 3 = 2x + 7$

$2x = 10 \Rightarrow x = 5$.

Side length $= 4(5) - 3 = 17$ (check: $2(5) + 7 = 17$ ✓).

Perimeter $= 3 \times 17 = 51$.

Answer: $x = 5$; each side $= 17$; perimeter $= 51$.

Example 8: Roof truss word problem

A gable roof truss is shaped like an isosceles triangle. Each base angle measures x, and the peak (vertex) angle measures $40°$ less than twice a base angle. Find all three angles.

Solution

Translate: vertex angle $= 2x - 40$.

Apply the Angle-Sum Theorem: $x + x + (2x - 40) = 180$

$4x - 40 = 180$

$4x = 220 \Rightarrow x = 55$.

Base angles $= 55°$ each; vertex $= 2(55) - 40 = 70°$.

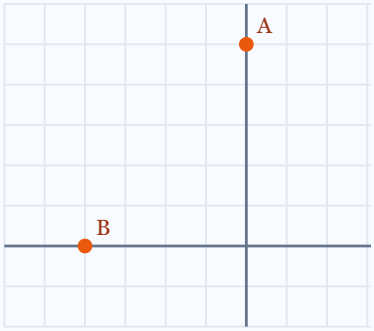
Check: $55 + 55 + 70 = 180$ ✓.

Answer: Base angles $55°$ each; vertex angle $70°$.

Example 9: Coordinate isosceles triangle

Triangle has vertices $A(0, 5)$, $B(-4, 0)$, $C(4, 0)$. Show it is isosceles, and state which angle is the vertex angle.

Solution



By symmetry about the y-axis, $AB = AC$.

Use the distance formula for the two legs:

$AB = \sqrt{(0 - (-4))^2 + (5 - 0)^2} = \sqrt{16 + 25} = \sqrt{41}$

$AC = \sqrt{(0 - 4)^2 + (5 - 0)^2} = \sqrt{16 + 25} = \sqrt{41}$.

Since $AB = AC = \sqrt{41}$, the triangle is isosceles with legs $\overline{AB}$ and $\overline{AC}$.

Answer: Vertex angle is $\angle A$ (between the congruent legs); $\angle B$ and $\angle C$ are the base angles.

Example 10: Vertex is a multiple of the base angle

In isosceles $\triangle ABC$, the vertex angle is 4 times each base angle. Find all three angles and classify by angles.

Solution

Let each base angle $= x$; then the vertex angle $= 4x$.

Apply the Angle-Sum Theorem: $x + x + 4x = 180$

$6x = 180 \Rightarrow x = 30$.

Base angles $= 30°$ each; vertex $= 4(30) = 120°$.

Check: $30 + 30 + 120 = 180$ ✓. Since the vertex angle exceeds $90°$, the triangle is obtuse.

Answer: Base angles $30°$ each; vertex $120°$ — obtuse isosceles triangle.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Base angles

Isosceles triangle, vertex angle $50°$. Find each base angle.

Answer: **Verification:**

Problem 2: Vertex angle

Isosceles triangle, base angles $62°$ each. Find the vertex angle.

Answer: **Verification:**

Problem 3: Converse

In $\triangle ABC$, $\angle B \cong \angle C$. What can you conclude about $\overline{AB}$ and $\overline{AC}$?

Answer: **Verification:**

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Problem 4: Equilateral angle

Find each angle of an equilateral triangle.

Answer: **Verification:**

Problem 5: Algebra — base angles

Base angles are $4x$ and $3x + 12$. Find x and each base angle.

Answer: **Verification:**

Problem 6: Algebra — vertex and base

Vertex angle is x, base angles are $2x$ each. Find x and all three angles.

Answer: **Verification:**

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Problem 7: Perimeter

Equilateral triangle has side length 9. Find the perimeter.

Answer: **Verification:**

Problem 8: Identify base angles

$AB = AC$ in $\triangle ABC$. Which two angles are the base angles?

Answer: **Verification:**

Problem 9: Base angle from vertex

Isosceles triangle, vertex angle $110°$. Find each base angle.

Answer: **Verification:**

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Problem 10: Equiangular

An equiangular triangle is also what type of triangle by sides?

Answer: **Verification:**

Problem 11: Algebra — base angles

Base angles are $5x - 4$ and $3x + 10$. Find x, each base angle, and the vertex angle.

Answer: **Verification:**

Problem 12: Vertex from base

Isosceles triangle, base angles $48°$ each. Find the vertex angle.

Answer: **Verification:**

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Problem 13: Challenge — equilateral algebra

An equilateral triangle has two sides represented by $4x - 3$ and $2x + 7$. Find x, the side length, and the perimeter.

Answer: **Verification:**

Problem 14: Challenge — coordinate isosceles

Vertices $A(0, 6)$, $B(-8, 0)$, $C(8, 0)$. Find all three side lengths, identify the vertex angle and base angles, and find the perimeter.

Answer: **Verification:**

Problem 15: Challenge — truss word problem

An isosceles roof truss has each base angle $= x$ and vertex angle $= 3x - 10$. Find x, all three angles, and classify by angles.

Answer: **Verification:**

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✓ Answer Key

1. $65°$
2. $56°$
3. $\overline{AB} \cong \overline{AC}$
4. $60°$
5. $x = 12$; base angles $48°$ each
6. $x = 36$; angles $36°$, $72°$, $72°$
7. 27
8. $\angle B$ and $\angle C$
9. $35°$
10. Equilateral
11. $x = 7$; base angles $31°$ each; vertex $118°$
12. $84°$
13. $x = 5$; side $= 17$; perimeter $= 51$
14. $AB = AC = 10$, $BC = 16$; vertex angle at A, base angles at B and C; perimeter $= 36$
15. $x = 38$; base angles $38°$ each, vertex $104°$ — obtuse isosceles

Great work — keep practising!